

# Venu Gopal Reddy Beereddy

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## Summary

**Associate Engineer** focused on building intelligent systems using **Python**, DSA, and cutting-edge LLM frameworks. Proven experience implementing **Retrieval Augmented Generation (RAG)** architectures with **LangChain** and **ChromaDB** to solve real-world business challenges and deliver measurable outcomes.

## Technical Skills

**Frontend:** React.js, Redux, React Router, HTML, CSS

**Backend:** Node.js, Express.js, REST APIs

**Databases:** MongoDB, MySQL

**AI/ML/Data:** LangChain, ChromaDB, HuggingFace, RAG, ML Models (Logistic Regression, SVM)

**Tools:** Git, Swagger, Postman, Vercel, Render

**Languages:** Java, JavaScript, Python, SQL

## Projects & Experience

### Real-Time Auction Platform

*React.js, Node.js, Express.js, MongoDB*

GitHub | Live Demo

- Directed a **5-member team** to design and deploy a scalable auction platform with role-based access, supporting **100+ secure, real-time transactions**.
- Designed and documented **RESTful APIs** with Node.js/Express.js and MongoDB, ensuring smooth integration with the React frontend.
- Implemented OTP-based **multi-factor authentication**, reducing fraudulent sign-ups by **40%**.
- Optimized MongoDB indexing, increasing backend query performance and overall platform responsiveness by **35%**.

### AI-Powered PDF Chatbot Project

*Streamlit, LangChain, ChromaDB, HuggingFace*

GitHub

- Developed an interactive AI-powered PDF Chatbot using **Streamlit** for the frontend, enabling users to upload multiple documents and query their content in natural language.
- Engineered the data pipeline for document retrieval, employing **PyPDF2** for text extraction, chunking large texts for optimized search, and utilizing **LangChain** for orchestration.
- Implemented a custom **Retrieval Augmented Generation (RAG)** system by generating vector embeddings with **HuggingFace BGE** models and persisting the knowledge base in **ChromaDB**.
- Delivered a conversational experience with **memory** integration, allowing the chatbot to maintain context across multiple turns and providing real-time, accurate answers grounded in the uploaded PDFs.

### Bitcoin Price Prediction System

*Python, Flask, Logistic Regression, SVM, Decision Tree*

GitHub

- Built from-scratch **ML models** (Logistic Regression, SVM, Decision Tree) delivering **85% predictive accuracy** in forecasting Bitcoin price movements without relying on external ML libraries.
- Developed a **Flask-based dashboard** for real-time inputs and instant prediction visualization, enabling faster data-driven user insights.
- Constructed robust **data preprocessing pipelines** (feature extraction, normalization, temporal transforms) that improved test dataset reliability by **20%**.

## Education

**Indian Institute of Information Technology, Sri City, India** *Nov 2022 – Jun 2026 (Expected)*

Bachelor of Technology in Computer Science

**Relevant Courses:** Data Structures and Algorithms, Operating Systems, Database Systems, Computer Networks, Computer Architecture

## Certifications

- Fullstack Web Development Bootcamp – GeeksforGeeks** – Covered HTML, CSS, JavaScript, React, Node.js, Express, and MongoDB.

## Extracurricular Activities

- Coordinated **8 volunteers** for a Swachh Bharat drive, engaging **200+ students** in community service and campus clean-up.
- Managed logistics and communication for a **Mega Marathon with 200+ participants**, ensuring smooth event execution and strong student involvement.

## Achievements

- Solved **100+ Data Structures & Algorithms problems on LeetCode**, improving coding speed and accuracy, and strengthening algorithmic expertise.